

Experiment Report: Checkout Button Colour Test

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Recommendation: SHIP

Confidence: HIGH | Risk: LOW

- P(treatment best) = 98.9% exceeds 95% threshold
- Frequentist test significant at alpha=0.05 (p=0.0204)
- Relative lift of 14.8% is meaningful
- Expected annual revenue uplift: \$303,862
- Expected value of shipping: \$814.62/day vs keeping control: \$0.00/day

Experiment design

Field	Value
Hypothesis	Changing the CTA button from grey to green increases conversion rate
Variants	Control (Grey Button) vs Treatment (Green Button)
Sample size	10,000 control / 10,000 treatment
Duration	14 days
Significance level	alpha = 0.05

Frequentist results

Metric	Control	Treatment	Lift
Conversion rate	5.01%	5.75%	+14.77% rel.
95% CI	[4.60%, 5.46%]	[5.31%, 6.22%]	
Revenue / user	\$2.532	\$2.839	+12.14% rel.

Two-proportion z-test: $z = 2.319$, $p = 0.0204$ (significant at $\alpha = 0.05$). Welch t-test (revenue/user): $t = 1.887$, $p = 0.0591$, Cohen's $d = 0.0267$. Power: 35.5% for a 10% MDE (required $n = 31,152/\text{group}$).

Bayesian results

Metric	Value
P(treatment > control)	98.9%
Expected relative uplift	+14.74%
Expected loss if shipping treatment	0.0011 pp
Expected loss if keeping control	0.7394 pp
Control 95% credible interval	[4.60%, 5.46%]
Treatment 95% credible interval	[5.31%, 6.22%]

Prior: Beta(1, 1)

Business impact

Assumptions: 5,000 daily visitors, 50% of traffic affected, \$45.00 revenue per conversion.

Metric	Value
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Daily conversion lift	+18.5 conversions/day
Daily revenue lift	+\$832.50/day
Annual revenue lift (point estimate)	+\$303,862
Annual range (treatment CI)	\$123,390 to \$498,290
EV of shipping vs status quo	+\$814.62/day

Scenario analysis

Scenario	Assumed lift	Daily revenue impact	Annual revenue impact	Probability	Recommendatio
Null (no effect)	+0.000%	\$0.00	\$0	5%	Do not ship
Pessimistic	+0.300%	\$338.05	\$123,390	10%	Ship treatment
Base case	+0.740%	\$832.50	\$303,863	65%	Ship treatment
Optimistic	+1.213%	\$1,365.18	\$498,290	20%	Ship treatment

Caveats

- Annualised figures extrapolate a short test window; seasonality and novelty effects may erode the lift.
- Revenue impact assumes the conversion lift is the only effect (no change in average order value).
- Scenario probabilities are illustrative weights, not posterior probabilities.